

Moein Razavi

Senior iOS Engineer · Swift / SwiftUI / UIKit · Computer Vision & Wearable Health

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Professional Summary

Senior iOS Engineer with **5+ years** shipping production Swift apps for **iPhone and iPad**, including App Store and **enterprise distribution**. Built and maintained an in-house **computer vision SDK** (Swift + **Apple Vision**, real-time video processing, **Core ML**) and **wearable-data integrations** (CoreBluetooth, **HealthKit**, heart-rate/HRV pipelines). Led a complete **UIKit-to-SwiftUI design system rollout**, refactoring legacy UIKit into **modular, testable SwiftUI components**. Architected **CI/CD** with **Xcode Cloud**, Fastlane, GitHub Actions, and **TestFlight**. Built a secure **QR-based cross-device login** flow with **OAuth 2.0 / JWT token exchange**. Proven in **performance optimization**—memory profiling, frame rate tuning, launch time reduction, and latency optimization. Backend fluency in Python (Flask, FastAPI), TypeScript/Node.js, Firebase, and Supabase.

Technical Skills

iOS & Apple Frameworks: Swift (Expert), SwiftUI, UIKit, Combine, Swift Concurrency (async/await), Core ML, Apple Vision, AVFoundation, CoreBluetooth, HealthKit, Swift Charts, MapKit, Core Data, CloudKit, Core Animation, Grand Central Dispatch (GCD)

Architecture & Design Patterns: Model-View-ViewModel (MVVM), MVVM-Coordinator, Clean Architecture, Protocol-Oriented Programming (POP), Dependency Injection, Coordinator Pattern, UIKit/SwiftUI Bridging, Modular Components, Offline-First

CI/CD & Quality Assurance: Xcode Cloud, Fastlane, GitHub Actions, TestFlight, App Store Connect API, Enterprise Distribution, XCTest, XCUITest, Snapshot Testing, Code Coverage, Code Review Standards, Automated Code Signing

Computer Vision & ML: Apple Vision Framework, Core ML, Real-Time Video Processing, Object Detection (YOLO), On-Device ML Inference, OpenCV, TensorFlow, PyTorch, coremltools

Authentication & Security: OAuth 2.0, JSON Web Token (JWT), Token Exchange, QR-Based Cross-Device Login, Sign in with Apple, Firebase Auth, Keychain Services, Biometric Authentication (Face ID / Touch ID), Certificate Pinning

Wearable & Health Data: HealthKit, CoreBluetooth (BLE), Heart Rate / HRV Pipelines, Biosignal Processing, Wearable Sensor Fusion (Accelerometer, Gyroscope, PPG), Real-Time Streaming, Apple Watch

Backend & APIs: Flask, FastAPI, Node.js, Firebase (Auth, Firestore, FCM), Supabase (PostgreSQL, Auth, Storage), REST, WebSocket, GraphQL, MongoDB, Redis, Docker

Performance & Animation: Memory Profiling (Instruments), FPS / Frame Rate Optimization, Launch Time Reduction, Latency Reduction, Motion and Animation Polish, Image/Asset Caching, Concurrent Transfer Pipelines

Programming Languages: Swift (Expert), Python, TypeScript, JavaScript, C++, Java, SQL

Professional Experience

Software Engineer — iOS & AI/ML

Jun 2023 – Sep 2024

CereVu Medical, Inc. San Francisco Bay Area, CA

- Architected **2 SwiftUI iOS apps** (Vitality + Clinical, iPhone + iPad) with a shared **design system** (colors, typography, spacing, animation presets) and **CoreBluetooth** hardware integration (Nordic UART BLE) for real-time biosignal acquisition from medical wearable patches, charting **9 biomarkers** via Swift Charts.
- Led **100% UIKit-to-SwiftUI design system rollout**: planned and executed the full migrate of the Clinical App, **bridging** legacy UIKit and SwiftUI and **refactoring** into **modular, testable SwiftUI components** with MVVM-Coordinator, role-based views (Patient, Physician, Admin), iCloud sync, and CSV export, expanding data accessibility by **25%**.
- Developed a **computer vision SDK** embedding the ROPA pain-detection algorithm with on-device ML inference via **Core ML** and a C++ signal-processing bridge, processing **real-time video** sensor data at **60 FPS** with optimized **memory** footprint; extended **test coverage** with XCTest unit and UI tests, elevating monitoring precision by **30%**.
- Implemented **Sign in with Apple + OAuth 2.0 / JWT token exchange** with secure storage (**Keychain Services**), **biometric authentication** (Face ID), and role-based access control across patient and clinician workflows.
- Owned the full **iOS release pipeline**: CI/CD, **TestFlight**, and public/**enterprise distribution**; built the companion **Flask REST API** backend (MongoDB time-series, Redis caching, PyJWT Apple token verification) deployed via **Docker** on AWS EC2 with **GitHub Actions**, Gunicorn, and Nginx.
- Paired with design on **motion and animation polish** (spring animations, scale/opacity transitions) and with backend on **REST** and **GraphQL** API contracts; participated in **code review** standards to maintain quality across the team.

Aug 2023 – Present

Senior Data Scientist & AI/ML Engineer

Ford Motor Company Detroit, MI

- Designed an **AI-powered data ingestion agent** (Vertex AI Gemini) with two-stage architecture, replacing ~500 lines of utility-specific parsers with a single universal pipeline across CSV, Excel, PDF, XML, and HTML formats.
- Architected a **multi-agent AI system** with LangChain, RAG, and vector databases serving EV analytics across **10 US states**; built **FastAPI** microservices on Cloud Run with **OAuth 2.0**, BigQuery, and Tekton CI/CD.

Sep 2021 – May 2024

Research Assistant & ML Engineer

Texas A&M University College Station, TX

- Designed a dual-branch transformer for **wearable** stress detection from **Apple Watch heart-rate and HRV data (256 participants, 1.7B sensor rows)**, achieving **AUC 0.993**; deployed on-device via **Core ML** with **60% faster inference**, building a full **HealthKit**-ready real-time inference pipeline in Swift.
- Deployed a **computer vision** system for **real-time video processing**—physical distancing and mask detection using deep learning object detection—achieving **98% accuracy** monitoring **50,000+ daily interactions**.
- Developed an iOS app (GlucoseCoach) with Flask backend for real-time personalized glucose monitoring, applying explainable AI (SHAP, LIME) for clinical decision support.
- Created **OpenSync**—an open-source real-time synchronization platform for multi-modal **wearable** sensor data (EEG, eye-tracking, biosensors) via WebSocket pipelines, adopted by **3 research labs** (31 citations).

Freelance Projects

Jan 2023 – Present

Clink Social — SwiftUI Social Platform

Freelance iOS Development

- Built Clink Social from scratch as a **SwiftUI** social platform with **Firebase** (Auth, Firestore, Storage, FCM), real-time messaging via **Combine**, MapKit discovery, and fan-out timeline architecture, increasing user engagement by **22%**.
- Implemented a secure **QR-based cross-device** sign-in flow: generate a time-boxed QR code in the mobile app, scan on a second device, exchange tokens with the API via **OAuth 2.0 / JWT**, and log users in instantly, reducing sign-in friction by **35%**.
- Developed TypeScript/Node.js Cloud Functions for feed aggregation, push notifications, and content moderation, reducing inappropriate content by **30%**.

Jan 2023 – Present

Inspecalytics — Full-Stack Field Inspection Platform (iOS + Web)

Freelance Full-Stack Development

- Built a **cross-platform inspection platform**—SwiftUI iOS app (iPhone + iPad) and companion web app—sharing a unified **Supabase** backend for seamless data sync; engineered custom **AVFoundation** camera with annotation overlays and in-app **DOCX/PPTX/XLSX report generation**, slashing report time by **35%**.
- Implemented **offline-first architecture** with SHA-256 change detection, debounced cloud sync, and concurrent image transfers; automated **Fastlane + GitHub Actions** CI/CD for **TestFlight**, cutting deployment time by **40%**.

Education

Aug 2021 – Dec 2023

Ph.D., Industrial Engineering (Data Science) | GPA: 4.0/4.0 | Dean's List | Research Fellowship

Texas A&M University, College Station, TX

Published **4 peer-reviewed papers** (IEEE, JMIR, SN Computer Science). *Focus*: Computer Vision, Wearable ML, Health Data.

Aug 2019 – May 2021

M.S., Computer Science | GPA: 4.0/4.0 | Dean's List | Research Fellowship

Texas A&M University, College Station, TX

Coursework: Deep Learning, Computer Vision, NLP, Algorithm Design, Database Systems, Software Engineering.

Certifications

Deep Learning Specialization – DeepLearning.AI · **Machine Learning Specialization** – Stanford University · **NLP Specialization** – DeepLearning.AI · **Generative AI with LLMs** – AWS